MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, ANALOG/DIGITAL, DIGITAL, KEY CODE QQQ

This amendment forms a part of Military Specification MIL-M-28787/232A(EC), dated 21 December 1981, and is approved for use by the Naval Electronic Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 6

* TABLE I, Test condition M1, M2, Max columns: Delete "15" (four places) and substitute "20" (four places).

PAGE 7

* TABLE I, Test condition X1-X4, Operating temperature 0° to 60°C Under Min columns delete "-4.88" (two places) and substitute "-9.76" (two places) and under Max columns delete "4.88" (two places) and substitute "9.76" (two places).

PAGE 8

6.1, delete and substitute.

"6.1 Definitions. Definitions are as specified in MIL-STD-1331 and as specified below.

"6.1.1 Resolution. Resolution is the smallest change (ideally 1 LSB) in the analog input voltage that results in a change in the digital output code. The resolution of a converter is dependent upon the number of digital bits that make up the converter and the input analog full scale voltage range.

"6.1.2 Zero error. Zero error is the center point of the analog input voltage range required to obtain a digital output code which corresponds to a zero analog input voltage

"6 1 3 Linearity error. Linearity error is the deviation from a straight line drawn through the end points of the plotted values of the converter's transfer characteristic. This specification puts a limit on the error due to imperfect bit weighting, such that the sum of either positive or negative errors of the individual bits does not exceed the limit.

"6.1.4 Differential nonlinearity. Differential nonlinearity is the deviation of an actual bit size from the theoretical value (1 LSB) over the full range of the digital output. This specification limits the analog input voltage step between any two adjacent digital codes over the full range of the digital output.

"6.1.5 Monotonicity. Monotonicity is the behavior in a converter that guarantees that the slope of the transfer characteristic of the converter does not change polarity in response to a continuously increasing (or decreasing) analog input voltage.

"6.1.6 Absolute error (%full scale). Absolute error is the deviation between the actual and the full-scale values of the converter's transfer characteristic.

"6.1.7 Hysteresis. Hysteresis is the variation in the analog input voltage range that will produce the same digital output code (ideally 1 LSB) as that code is approached from its two adjacent codes."

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PAGE 9

6.4, Insert the following between Digital input voltage and Resolution:

" Digital output drive current

Low level (V_{OL} = 0.3 V)	3.2 mA minimum
High level (V_{OH} = 2.4 V)	-80 µA minimum "

* 6.4, Zero error: Delete "±1LSB maximum" and substitute "±2LSB maximum."

PAGE 19

FIGURE 6 Delete and substitute new figure 6 as printed on page 3 of this amendment.

PAGES 25 AND 26

FIGURE 9 Delete and substitute new figure 9 as printed on pages 4 and 5 of this amendment.

PAGES 49 AND 50

FIGURE 17 Delete and substitute new figure 17 as printed on pages 6 and 7 of this amendment.

PAGES 52 AND 53

FIGURE 18 Delete and substitute new figure 18 as printed on pages 8 and 9 of this amendment.

PAGE 54

* TABLE XIV, Test conditions M1 and M2, Max column Delete "15" (two places) and substitute "20" (two places).

PAGE 76

* TABLE XX, Test conditions T1 through T4 Under Min column delete "-3.3398", "3.3252", "3.3252", and "-3.3398" and substitute "-3.410", "3.255", "3.255", and "-3.410" respectively, Under Max column delete "-3.3202", "3.3447", "3.3447", and "-3.3203" and substitute "-3.250", "3.415", "3.415", and "-3.250" respectively.

PAGES 84, 85, 86, AND 87

FIGURE 34 Delete and substitute new figure 34 as printed on pages 10, 11, 12, and 13 of this amendment.

PAGES 93, 94, AND 95

* FIGURE 38 Delete and substitute new figure 38 as printed on pages 14, 15, and 16 of this amendment.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 5963-N465)

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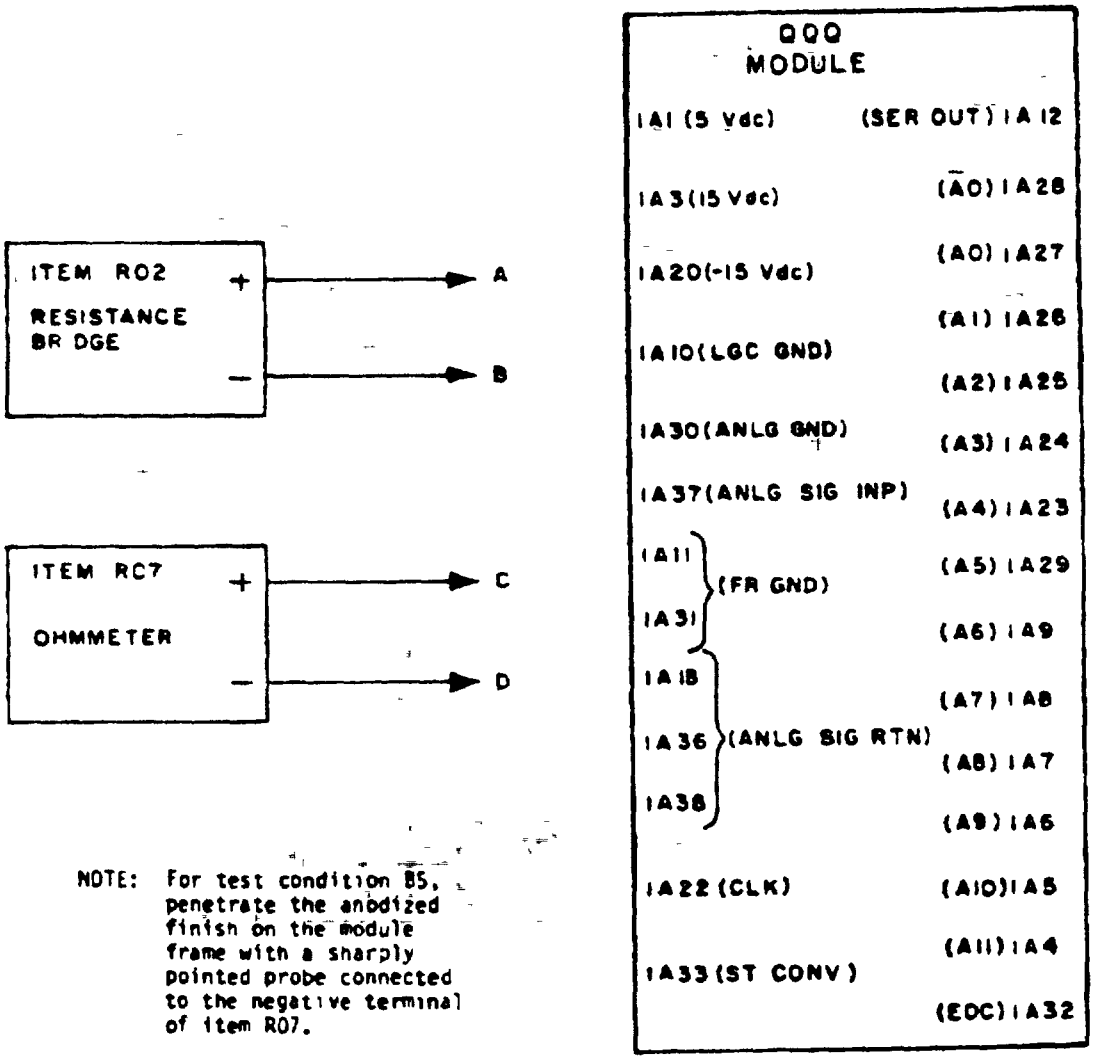


FIGURE 6. Continuity and isolation test circuit for tables IV and V.

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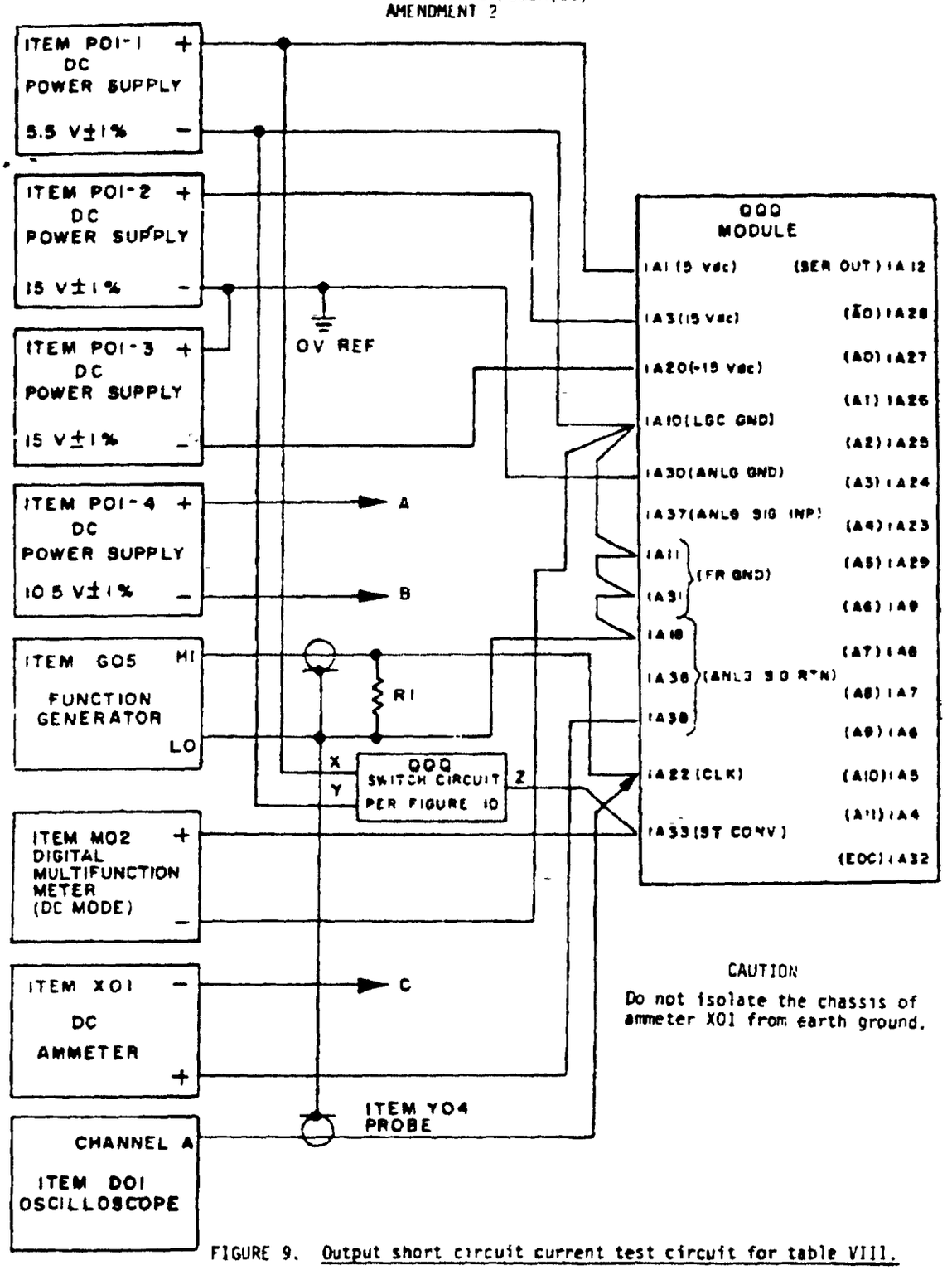


FIGURE 9. Output short circuit current test circuit for table VIII.

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NOTES
1. R1 = 51 ±5%, 1/2 W
2 When performing the output short circuit test only one output shall be shorted at a time The duration of each test shall not exceed 5 seconds
3. Item G05 setup
   Square wave output
   Frequency = 200 kHz ±25 .
   High level = 4.0 ±1.0 V.
   Low level = 0.0 ±0.2 V.
4 Test procedure for test condition F1·

   a. Connect the test circuit equipment as specified (do not connect point C at this time) for the appropriate test condition
   b. Set switch S1 of figure 10 to position 2 and allow at least one low-to-high clock pulse at pin 22. The voltage reading on item M02 shall be 0 0 to 0.4 V
   c. Set switch S1 of figure 10 to position 1 and allow at least 12 low-to-high clock pulses at pin 22 The voltage reading on item M02 shall be 2.4 to 5 5 V
   d Connect point C as specified in table VIII The current measured on ammeter X01 shall satisfy the limits of table VIII

5 Test procedure for test conditions F2 - F14

   a Repeat steps 4a - 4d Steps 4b and 4c are only required once for test conditions F2 - F14.

6 Test procedure for test condition F15

   a Repeat steps 4a, 4b, and 4d.

FIGURE 9. Output short circuit current test circuit for table VIII - Continued

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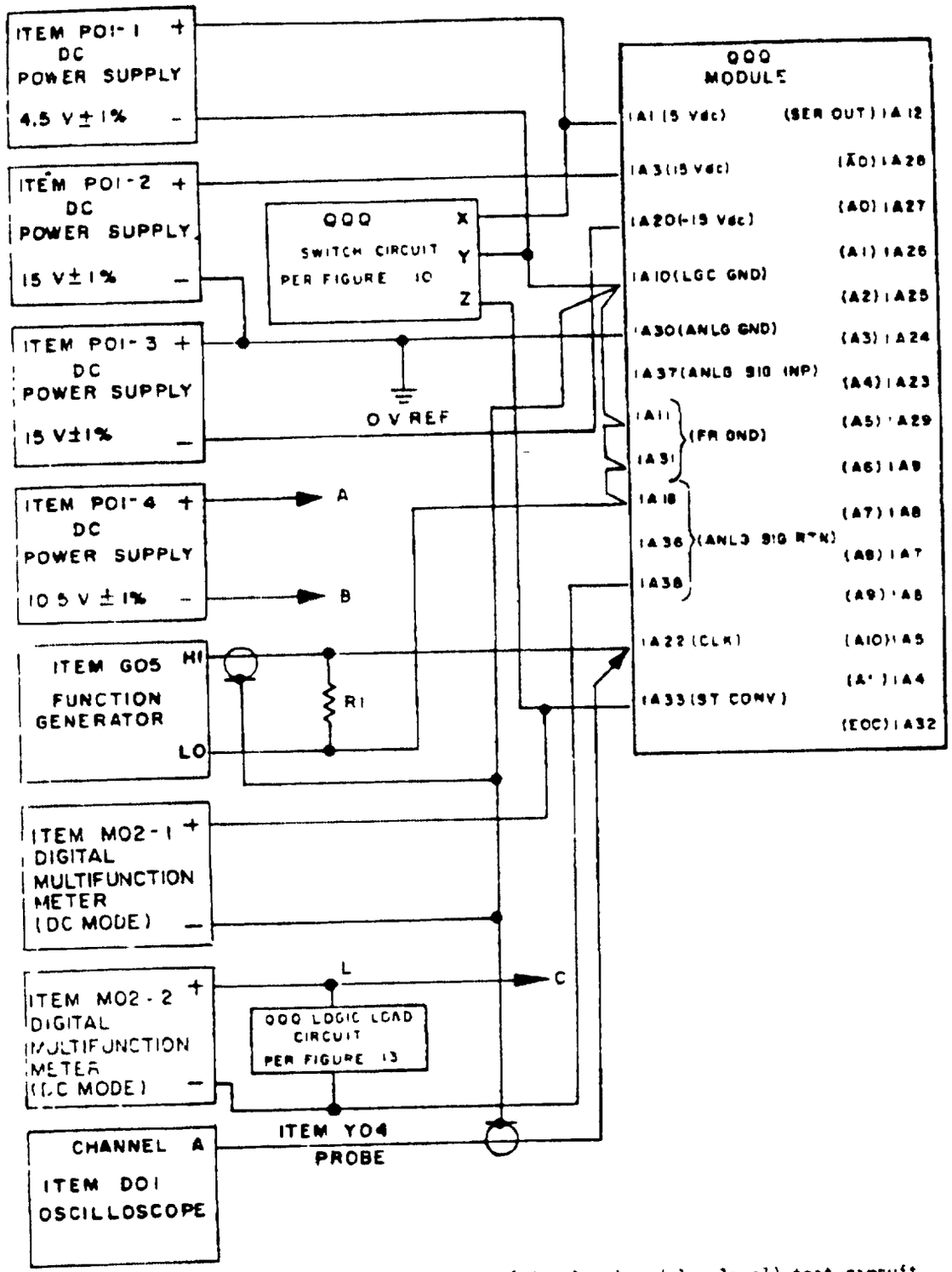


FIGURE 17 Static output voltage (high-level and low-level) test circuit for table XII.

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NOTES
1. R1 = 510 ±5%, 1/2 W
2 Item G05 setup
  Square wave output.
  Frequency = 200 kHz ±25%
  High level = 4.0 ±1.0 V
  Low level = 0.0 ±0.2 V.
3 Since items MO2-1 and MO2-2 are not used simultaneously, only one item MO2 is required.
4. Test procedure for test conditions K1-K15.

  a Connect the equipment as specified for the appropriate test condition.
  b Steps 4c and 4d are only required once for test conditions K1-K15
  c. Set switch S1 of figure 10 to position 2 and allow at least one low-to-high clock pulse at pin 22 The voltage reading on item MO2-1 shall be 0.0 to [illegible]
  d Set switch S1 of figure 10 to position 1 and allow at least 12 low-to-high clock pulses at pin 22 The voltage reading on item MO2-1 shall be [illegible]
  e. The voltage measured on item MO2-2 shall satisfy the limits of table XII

5 Test procedure for test conditions K16-K29 Repeat steps 4a - 4d.
6 Test procedure for test condition K30. Repeat steps 4a, 4c, and 4e

FIGURE 17. Static output voltage (high-level and low-level) test circuit for table XII - Continued.

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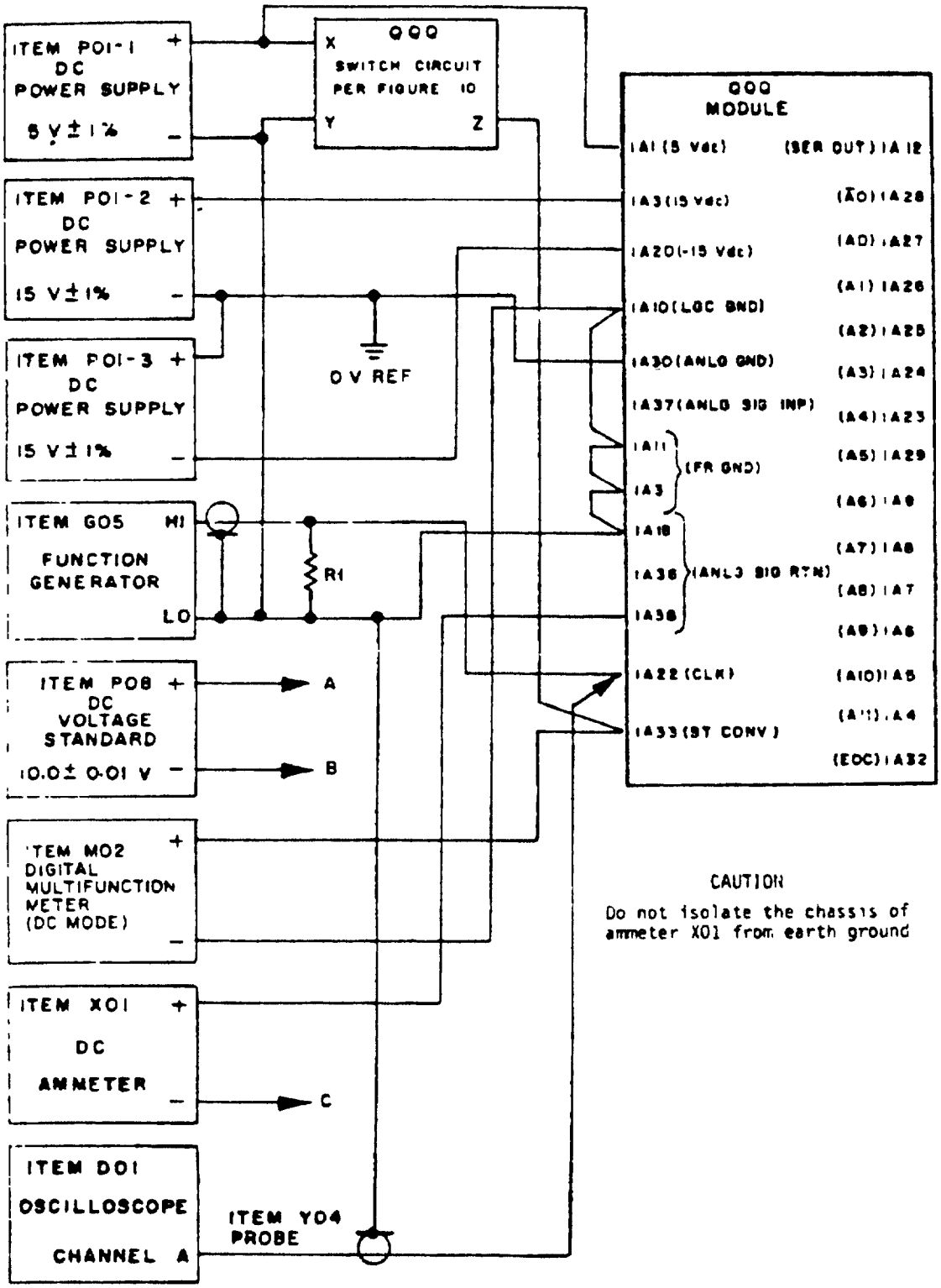


FIGURE 18. Input impedance check test circuit for table XIII.

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NOTES
1 R1 = 51Ω ±5%, 1/2 W.
2 Trem G05 setup
Square wave output
Frequency = 200 kHz ±25%
High level = 4.0 ±1.0 V
Low level = 0.0 ±0.2 V
3 Test procedure for test conditions L1 and L2

a Connect the equipment as specified for the appropriate test condition
b Set switch S1 of figure 10 to position 2 and allow at least one low-to-high clock pulse at pin 22. The voltage reading on item M02 shall be 0.0 to 0.4 V
c Set switch S1 of figure 10 to position 1 and allow at least 12 low-to-high clock pulses at pin 22. The voltage reading on item M02 shall be 2.4 to 5.0 V
d The current measured on item X01 shall satisfy the limits of table XIII

FIGURE 18. Input impedance check test circuit for table XIII - Continued.

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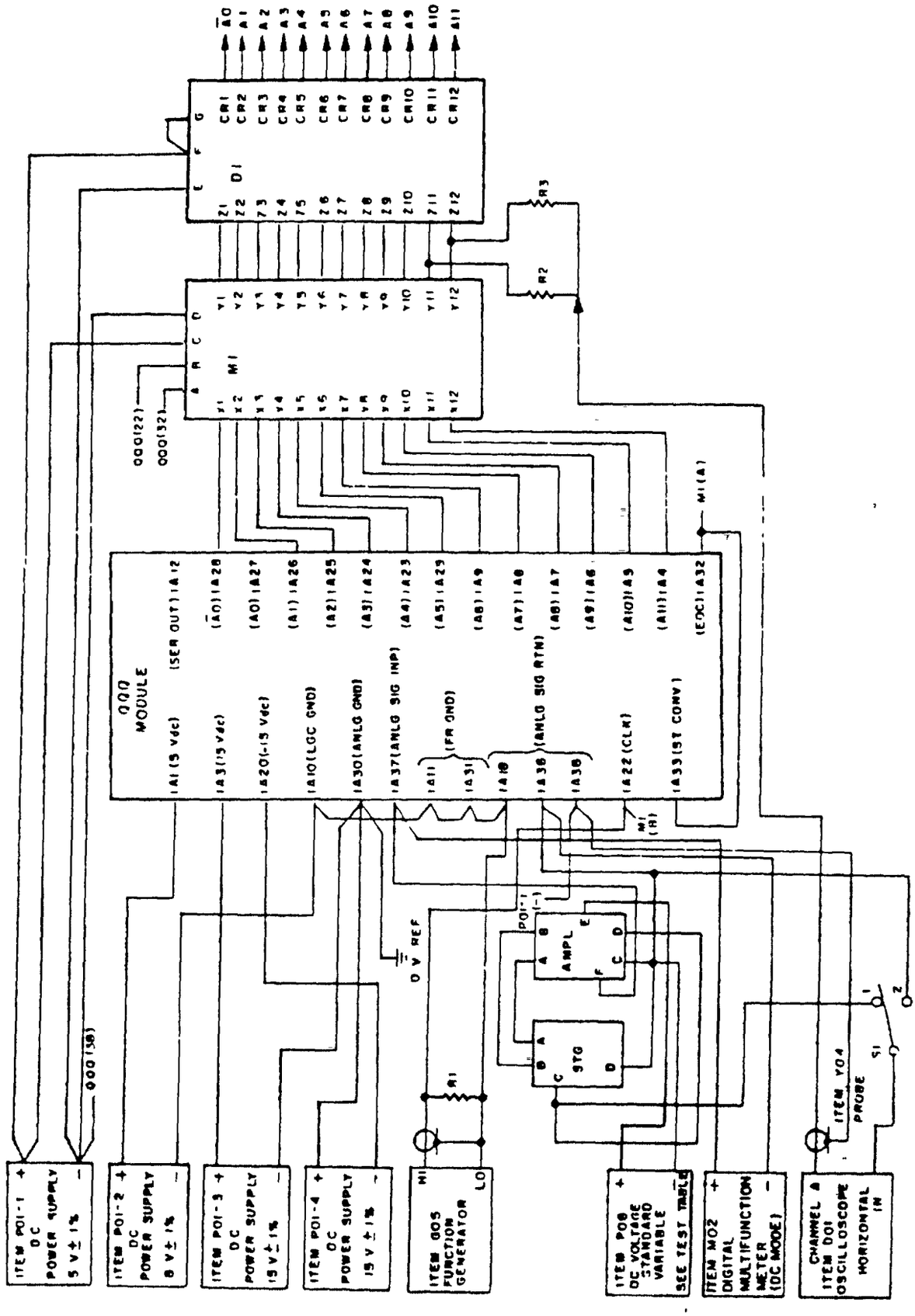


FIGURE 34 Hysteresis, differential nonlinearity, and linearity test circuit for tables XXI, XXII, and XXIII.

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NOTES
1. M1 - Data register per figure 22
2. D1 - Level detector per figure 23.
3. AMPL - Summing amplifier per figure 36.
4. STG - Sawtooth dither generator per figure 35
5. R1 - 51Ω ±5%, 1/2 W
6. R2 - 4.99 kΩ ±5%, 1/4 W
7. P3 - 10 kΩ ±5%, 1/4 W.
8. S1 - SPDT.
9. Item G05 setup:
   Square wave output.
   Frequency = 330 kHz ±5%
   High level = 4.0 ±1.0 V.
   Low level = 0 ±0.2 V.
   Set parameters using item D01, referenced to pin 10.
10. Calibration procedure for test conditions U1 - U7.

    a. Set horizontal display of item D01 to EXT-X1 and the vertical sensitivity to 2 V/division.
    b. Set item P08 to zero volts and switches S1, S2, S3, and S4 to position 1. Adjust vertical position control until display is centered.
    c. Adjust horizontal variable control of item D01 until at least 50% of the steps displayed are 1 ±0.5 divisions wide
    d. Set switch S3 to position 2. The voltage reading on item M02 shall be 0.0 ±1.0 mV. If this criteria is not met, replace A1 of figure 36 until criteria is satisfied.
    e. Set switch S3 to position 1 and switch S1 to position 2.
    f. Adjust the horizontal position control of item D01 until display is alined (±0.1 division) with the center vertical graticule
    g. Set switch S1 to position 1. The display shall be as shown on figure 37.

11. Test conditions U1 - U7.

    a. Set switches S1, S2, and S4 to position 1 and switch S3 to position 2.
    b. Set item P08 per table XXI to obtain the digital output code, displayed by level detectors, specified for the appropriate test condition.
    c. Set switch S3 to position 1 and adjust item P08 until E7, defined in figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule. Record the setting of item P08.
    d. Set switches S2 and S3 to position 2 and repeat steps 11b thru 11c
    e. The difference in the recorded values of steps 11c and 11d shall satisfy the requirements of table XXI.
    f. Repeat steps 11a and 11b.
    g. Set switch S3 to position 1 and adjust item P08 until E2, defined in figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule. Record the setting of item P08.
    h. Set switches S2 and S3 to position 2 and repeat steps 11b and 11g
    i. The difference in the recorded values of steps 11g and 11h shall satisfy the limits of table XXI.

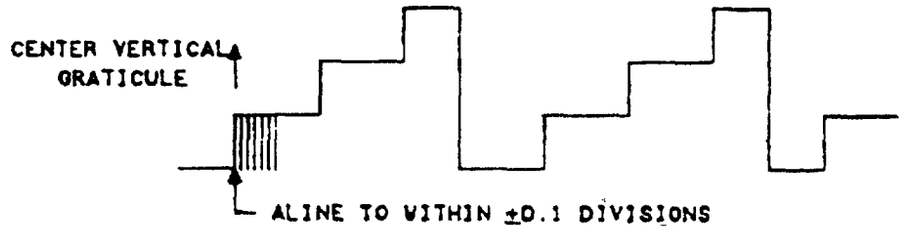
FIGURE 34 Hysteresis, differential nonlinearity, and linearity test circuit for tables XXI, XXII, and XXIII - Continued

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*12 Calibration procedure for test conditions V1-V26 and W1-W7, must be performed each time a new module is tested.

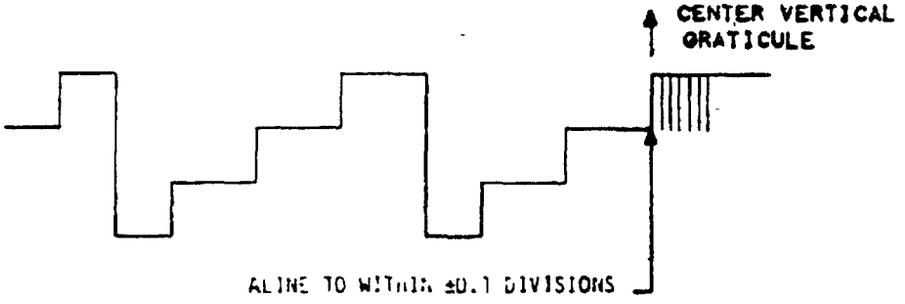
a. Set horizontal display of item D01 to EXT-X1 and the vertical sensitivity to 2 V/division
b. Set item P08 to zero volts and switches S1, S2, S3, and S4 to position 1. Adjust vertical position control until display is centered.
c. Adjust horizontal variable control of item D01 until at least 50% of the steps displayed are 1 ±0.5 divisions wide.
d. Set switch S3 to position 2. The voltage reading on item M02 shall be 0.0 ±1.0 mV. If this criteria is not met, replace A1 of figure 36 until this criteria is satisfied.

e. Set switch S3 to position 1 and switch S4 to position 2.
f. Aline E1 of the display, defined in figure 37, closest to the center vertical graticule with the graticule (±0.1 divisions) by adjusting the horizontal position control.
g. Set item P08 to +10 V and adjust R8 of figure 36 until the positive saturation display as defined below is obtained.



Positive saturation oscilloscope display

h Set item P08 to -9.9902 Vdc. Adjust the horizontal position control until the negative saturation display is alined as shown below.



i Repeat steps g and h at least once or as many times as required to obtain both alinements within ±0.1 divisions

Negative saturation oscilloscope display

i Repeat steps g and h at least once or as many times as required to obtain both alinements within ±0.1 divisions

FIGURE 34. Hysteresis, differential nonlinearity, and linearity test circuit for tables XXI, XXII, and XXIII - Continued

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13 Test conditions V1 - V13.

a Set switches S2, S3, and S4 to position 2 and switch S1 to position 1
b. Set item P08 per table XXII to obtain the digital output code (displayed by level detectors) specified for the appropriate test condition.
c Set switch S3 to position 1 Vary item P08 until the step leading edge as defined in table XXII and in figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule Record setting of item P08.
d. Adjust item P08 until the step trailing edge as defined in table XXII and in figure 37, closest to the center vertical graticule is alined (±0 1 divisions) with the graticule. Record the setting of item P08
e. The difference in the recorded values of steps 13c and 13d shall satisfy the limits of table XXII

14 Test conditions V14 - V26

a Set switches S1 and S2 to position 1 and switches S3 and S4 to position 2.
b Repeat steps 13b thru 13e

* 15 Test conditions W1 - W7.

a. Recalibrate test setup per steps 12a - 12i.
b Set switches S2, S3, and S4 to position 2 and switch S1 to position 1
c Set item P08 per table XXIII to obtain the digital output code (displayed by level detectors) specified for the appropriate test condition
d Set switch S3 to position 1. Vary item P08 until the trailing edge of the step to be measured, defined in figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule
e The settling of item P08 shall satisfy the limits of table XXIII.

FIGURE 34 Hysteresis, differential nonlinearity, and linearity test circuit for tables XXI, XXII, and XXIII - Continued

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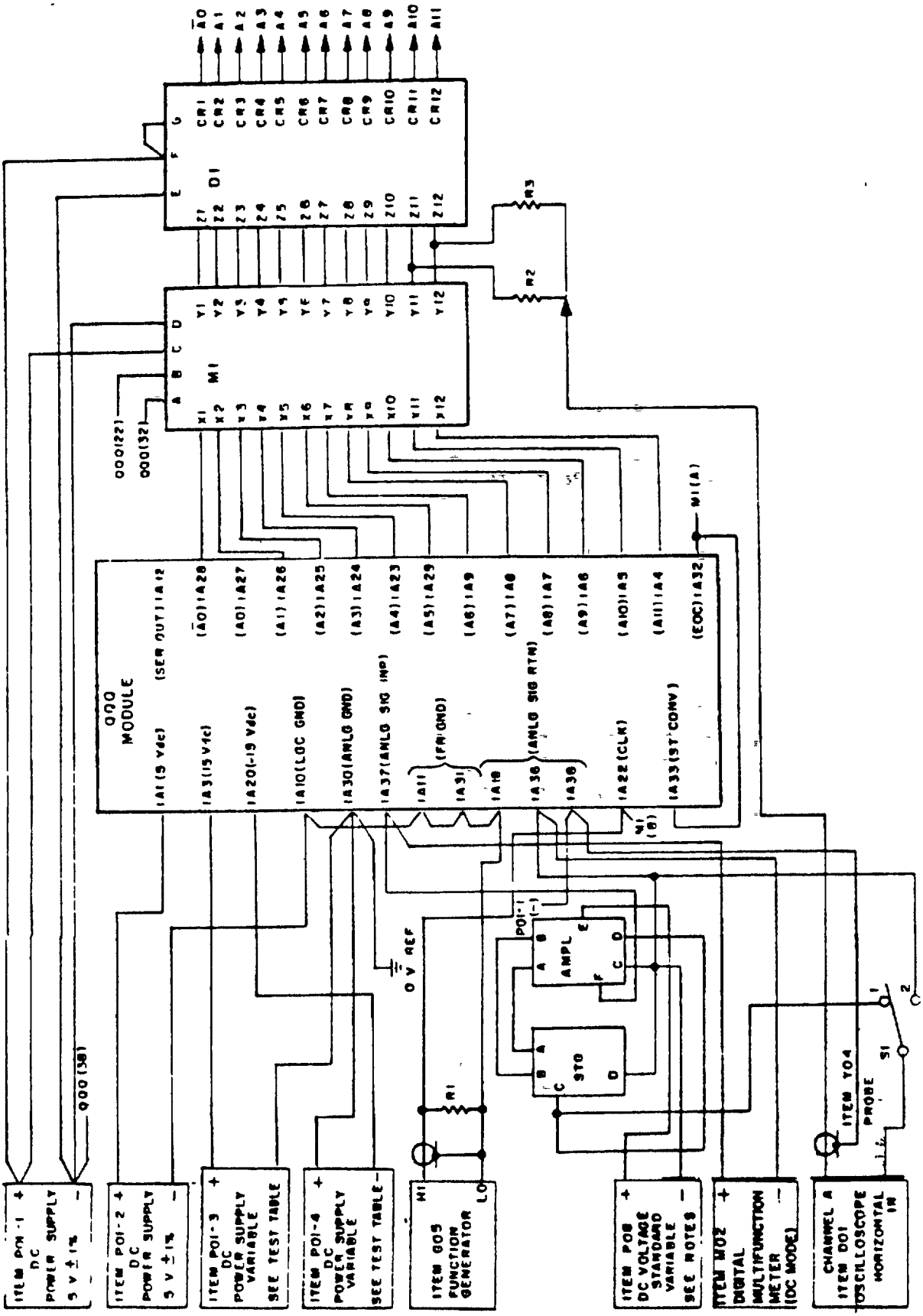


FIGURE 3R. Zero error and absolute error test circuit for tables XXIV and XXV.

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NOTES:
1. M1 = Data register per figure 22.
2. D1 = Level detector per figure 23.
3. AMPL = Summing amplifier per figure 36.
4. STG = Sawtooth dither generator per figure 35.
5. R1 = 51[illegible] ±5%, 1/2 W
6. R2 = 4.99 k[illegible] ±5%, 1/4 W.
7. R3 = 10 k[illegible] ±5%, 1/4 W.
8. S1 = SPDT switch
9. Item G05 setup:
   Square wave output.
   Frequency = 200 kHz ±10%.
   High level = 4.0 ±1.0 V.
   Low level = 0.0 ±0.2 V.
   Set parameters using item D01, referenced to pin 10.
10. Calibration procedure for test conditions X1 - X4 and Y1 - Y8.

    a. Set horizontal display of item D01 to EXT-X1 and the vertical sensitivity to 2 V/division
    b Set item P08 to zero volts and switches S1, S2, S3, and S4 to position 1. Adjust vertical position control to center the display.
    c. Adjust horizontal variable control of item D01 until at least 50% of the steps displayed are 1 ±0.5 divisions wide.
    d Set switch S3 to position 2. The voltage reading on item M02 shall be 0.0 ±1.0 mV. If this criteria is not met, replace A1 of figure 36 until criteria is satisfied.
    e Set switch S3 to position 1 and switch S1 to position 2.
    f. Adjust the horizontal position control of item D01 until the dot display is alined (±0.1 divisions) with the center vertical graticule.
    g. Set switch S1 to position 1. The display shall be as shown on figure 37.

11. Test conditions X1 - X4

    a. Set switches S2 and S3 to position 2 and switches S1 and S4 to position 1
    b Adjust item P08 until the voltage reading on item M02 is 0.0 ±0.1 mV. Record the setting of item P08
    c. Adjust item P08, 0.0 ±20 mV, to obtain the digital output code, displayed by level detectors, specified below.

$\overline{A0}$	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
L	L	L	L	L	L	L	L	L	L	L	L

    d. Set switch S3 to position 1 and adjust item P08 until "W", as defined on figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule. Record the setting of item P08.
    e. The difference in the recorded values of steps 11b and 11d shall satisfy the limits of table XXIV.

FIGURE 38. Zero error and absolute error test circuit for tables XXIV AND XXV - Continued.

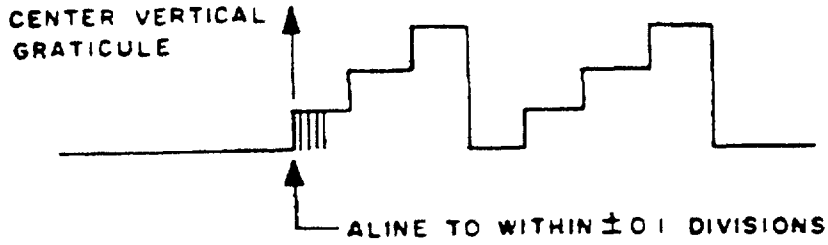
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12. Test conditions Y1 - Y4.

a. Set switches S2 and S3 to position 2 and switches S1 and S4 to position 1.
b. Adjust item P08 to 9.995 ±0.04 V to obtain the digital output code (displayed by level detectors) specified below.

A0	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
H	L	L	L	L	L	L	L	L	L	L	H

c. Set switch S3 to position 1 and adjust item P08 until a positive saturation oscilloscope display as defined below is obtained.



Positive saturation oscilloscope display

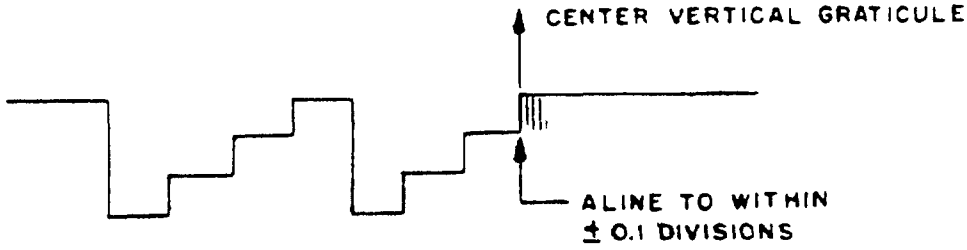
d. The setting of item P08 shall satisfy the limits of test conditions Y1 - Y4 of table XXV. If the digital output code cannot be obtained, the module shall be considered failed.

13. Test conditions Y5 - Y8.

a. Set switches S2 and S3 to position 2 and switches S1 and S4 to position 1.
b. Adjust P08 to -9.990 ±0.04 V to obtain the digital output code (displayed by level detectors) specified below.

A0	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
L	H	H	H	H	H	H	H	H	H	H	L

c. Set switch S3 to position 1 and adjust item P08 until a negative saturation oscilloscope display as defined below is obtained.



Negative saturation oscilloscope display

d. The settings of item P08 shall satisfy the limits of test conditions Y5 - Y8 of table XXV. If the digital output code cannot be obtained, the module shall be considered failed.

FIGURE 38. Zero error and absolute error test circuit for tables XXIV and XXV - Continued.

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